# Layer 7: SAM Chemistry

Mixed Probe-Thiol/MCH Monolayer: Deposition, Stability, and $Ca^{2+}$ Artefact

Eton iGEM 2026, Dry Lab | Layer 7 of 17-Layer Biosensor Pipeline | `sam_loading.py`, `sam_stability.py`, `calcium_artefact.py`

Layer 7 characterises the self-assembled monolayer (SAM) that immobilises the peptide probe on the gold electrode surface. Its primary output is the probe surface density $\Gamma_p$, which passes to Layer 8 (steric accessibility) and Layer 9 (surface kinetics). Secondary outputs are bounds on SAM degradation over the 60-minute assay window and a quantitative account of the $Ca^{2+}$ artefact and its ratiometric cancellation.

Every parameter value must be confirmed from a paper physically present in `papers/layer_07/`; values without a paper source are explicitly flagged throughout.

## I. Mixed SAM Architecture

**Molecular components**

The sensor electrode carries a two-component thiol monolayer. The probe-thiol consists of a peptide substrate bearing a methylene-blue (MB) redox label, tethered to gold via a $C_6$ alkyl spacer and a $PEG_6$ flexible linker (HS–$(CH_2)_6$–$PEG_6$–SGKIPRTLTA–MB). MMP-9 (or HNE on the secondary channel) cleaves the peptide, releasing the MB-bearing fragment into solution and producing a signal-off decrease in SWV peak current proportional to enzymatic activity (Layer 0, Section V).

The diluent is 6-mercapto-1-hexanol (MCH, HS–$(CH_2)_6$–OH). MCH fills gold sites not occupied by the probe-thiol, presenting a chemically inert surface to the wound fluid and preventing non-specific adsorption of wound proteins to the bare gold.

**Close-packed monolayer density**

The maximum surface density of a fully close-packed alkanethiol monolayer on gold, measured by reductive desorption coulometry, is [6]:

$$\Gamma_{\max} = 9.3 \times 10^{-10} \text{ mol cm}^{-2} = 930 \text{ pmol cm}^{-2} \tag{I.1}$$

Love et al. report an independent cross-check of approximately 750 pmol $cm^{-2}$ for densely packed alkanethiol monolayers on Au(111) [1]. Equation (**??**) is an upper bound. The probe-thiol cannot approach $\Gamma_{\max}$ due to the bulky peptide headgroup; the target probe coverage $\Gamma_p = 2$–$5$ pmol $cm^{-2}$ is 0.2–0.5% of the close-packed limit.

The following assumptions underlie this section:

1. The probe-thiol adsorbs to gold through the thiol–gold bond; MCH backfilling fills the remaining gold surface.
2. $\Gamma_{\max} = 930$ pmol $cm^{-2}$ is an upper bound on MCH packing; the actual MCH density is lower because probe-thiols occupy a fraction of the surface.

## II. Deposition Protocol and Target Surface Density

**DNA-thiol papers: inapplicable to peptide-thiol probe**

The papers in `papers/layer_07/` addressing DNA-thiol deposition (Herne 1997 [2], Steel 1998 [3], Petrovykh 2003 [4]) do not transfer to the peptide-thiol probe for the following reasons:

- **Herne 1997 [2]**: The 1.0 M $KH_2PO_4$ buffer and log-time deposition kinetics arise from electrostatic repulsion between the 25 ionisable phosphate groups per ssDNA strand. The SGKIPRTLTA peptide carries far fewer charges and does not require high ionic-strength buffer.
- **Steel 1998 [3]**: The functional coverage limit ($4 \times 10^{12}$ molecules $cm^{-2}$) was set by hybridisation steric hindrance. The relevant limit for enzyme cleavage is set by MMP-9 access, which is more stringent (Section **??**, below).
- **Petrovykh 2003 [4]**: Non-Langmuir adsorption kinetics apply to HS-ssDNA; peptide-thiol adsorption is not covered.

**Confirmed deposition protocol**

Wcisło et al. (2021) [11] describe immobilisation of a cysteine-terminated peptide substrate (SH–$(CH_2)_{10}$–CO–Arg-Val-Arg-Arg) on gold by the two-step procedure applicable to this device:

1. Immerse polished gold electrode in 5 mM degassed ethanolic solution of the peptide-thiol for 20 h at room temperature.

2. Rinse with ethanol and deionised water; dry under $N_2$.
3. Backfill with 2 mM ethanolic 1-octanethiol (analogous to MCH) for 60 min, to eliminate pinholes and block non-specific adsorption.
4. Optional protein block (BSA, 1% in PBS, 60 min at 4°C) if wound-matrix fouling is anticipated.

For this device, 1-octanethiol is replaced by MCH (6-mercapto-1-hexanol, HS–$(CH_2)_6$–OH) to match the $C_6$ alkyl chain length of the probe-thiol spacer. The target deposition concentration (5 mM) and time (20 h) are adopted from Wcisło 2021 [11]; exact optimisation for the specific SGKIPRTLTA–MB probe must be confirmed by measuring $\Gamma_p$ on the fabricated electrode.

**Confirmed functional coverage limits**

Adjémian et al. (2010) [12] characterised the relationship between surface coverage $\Gamma_0$ and cleavage efficiency for a ferrocene (Fc)-labelled tetrapeptide end-grafted on gold via a $PEG_7$ linker, using MCH backfill (architecture directly analogous to the MB-SGKIPRTLTA–$PEG_6$ probe). Their key finding (Fig. 2 and text):

- At saturated undiluted layers ($\Gamma_0 \approx 6 \times 10^{-11}$ mol cm$^{-2}$ = 60 pmol cm$^{-2}$), cleavage yield drops to ∼50% because chains form compact domains inaccessible to the enzyme.
- For $\Gamma_0 \leq 10^{-11}$ mol cm$^{-2}$ = 10 pmol cm$^{-2}$ with MCH backfill, cleavage yield exceeds 80%.
- The working coverage used throughout Adjémian 2010 is $(0.5–1) \times 10^{-11}$ mol cm$^{-2}$ = 5–10 pmol cm$^{-2}$.

The device target $\Gamma_p$ = 2–5 pmol cm$^{-2}$ therefore sits within the confirmed functional regime. Adjémian 2010 also confirms that surface kinetics follow the Michaelis–Menten two-step mechanism (Layer 9), with enzymatic constants measurable directly from the electrode response.

| **Parameter** | **Status** | **Source** |
|---|---|---|
| Deposition protocol | Confirmed (5 mM EtOH, 20 h, MCH backfill) | Wcisło 2021 [11] |
| Functional $\Gamma_p$ limit | Confirmed ($\leq$ 10 pmol cm$^{-2}$) | Adjémian 2010 [12] |
| $k_{cat}$, $K_m$ for SGKIPRTLTA | Confirmed (86.4 s$^{-1}$, 1.3 mM) | Kridel 2001 [10] |
| Absolute $\Gamma_p$ on fabricated electrode | **Unconfirmed**: must be measured | Coulometric integration |
| Surface $k_{eff,surf}$ | **Unconfirmed**: must be measured on device | Layer 9 |

Table 1: Status of probe-loading parameters. The two unconfirmed items require measurements on the fabricated electrode; they cannot be borrowed from solution or analogous probes.

**Probe spacing constraint from enzyme size**

Once $\Gamma_p$ is established from the literature, the average probe spacing is:

$$\ell = (\Gamma_p \cdot N_A)^{-1/2} \tag{II.1}$$

For the sensor to function, MMP-9 (molecular weight 92 kDa, hydrodynamic radius ≈ 3–4 nm) must approach and bind each surface-immobilised peptide. This imposes the requirement $\ell \gtrsim 2 \times r_{MMP\text{-}9} \approx 6–8$ nm, corresponding to $\Gamma_p \lesssim 3$ pmol cm$^{-2}$. At higher probe densities, neighbouring probe molecules sterically block enzyme access, reducing the effective cleavage rate and underestimating free MMP-9 activity. This constraint is more stringent than the hybridisation limit in Steel 1998 and must be confirmed with a cleavage-efficiency vs. coverage measurement.

The following assumptions underlie this section:

1. Probe-thiol adsorbs to gold via the thiol–gold bond; MCH backfilling fills remaining surface sites.
2. A target of $\Gamma_p \lesssim 3$ pmol cm$^{-2}$ is adopted on steric grounds (enzyme-access requirement); this value is not confirmed from any paper in `papers/layer_07/` and is flagged as Limitation 1.

## III. SAM Stability over the Assay Window

Three mechanisms can degrade the SAM during the 60-minute assay: oxidative desorption by $H_2O_2$, thiol exchange with free thiols in the wound matrix, and electrochemical desorption by the SWV scan potential.

**Oxidative desorption**

Schoenfisch and Pemberton characterised alkanethiol SAM oxidation under ambient conditions [5]. Their central finding is that ozone ($O_3$), not $O_2$ or UV, is the primary oxidant in air, converting Au–SR to $Au^0$ + $RSO_xH$. Under ambient $O_3$ at 28–60 ppb, a $C_{12}$ SAM on gold lost 36.9% of its total sulphur in 1 h (Table 5 of [5]).

In the present device, the oxidant is $H_2O_2$ at 10–100 $\mu$M in aqueous wound fluid; ozone is absent from the aqueous phase. The relative oxidising power of the two species is reflected in their standard reduction potentials:

$$E^\circ(O_3/O_2) = +2.07\text{ V}, \qquad E^\circ(H_2O_2/H_2O) = +1.76\text{ V} \tag{III.1}$$

The 310 mV thermodynamic advantage of $O_3$ corresponds to a rate advantage of several orders of magnitude for thiolate oxidation. The Schoenfisch data therefore provide a conservative upper bound: even under the harshest oxidative conditions tested (ambient $O_3$ in air), SAM loss is approximately 37% in 1 h for a $C_{12}$ chain. Aqueous $H_2O_2$ at wound-fluid concentrations is a substantially weaker oxidant; oxidative SAM loss over the 60-minute assay is not expected to approach this bound.

No quantitative rate constant $k_{ox}$ for $H_2O_2$ oxidation of thiol SAMs in solution is available in `papers/layer_07/`. The argument above is comparative only; see Limitation 2.

**Thiol exchange**

Schlenoff et al. investigated the exchange of adsorbed alkanethiols on gold by free thiol in solution [7]. The key finding is that the exchange rate shows "very little dependence of rate on thiol concentration," meaning the process is not bimolecular but involves a rate-limiting surface step. In hexane, the pseudo-first-order exchange rate constant is [7]:

$$k_1 = 1.7 \times 10^{-5}\text{ s}^{-1} \quad \text{(hexane, 22°C)} \tag{III.2}$$

The fraction exchanged at time $t$ is $f = 1 - \exp(-k_1 t)$. At $t = 3600$ s (60 min):

$$f = 1 - \exp(-1.7 \times 10^{-5} \times 3600) = 1 - e^{-0.061} \approx 5.9\% \tag{III.3}$$

Equation (**??**) is a strict upper bound: Schlenoff et al. further show that exchange in aqueous conditions is negligible, with essentially no exchange detected even after 125 h in water (Figures 3–6 of [7]). They conclude that desorption is "least pronounced in aqueous systems." Wound fluid is an aqueous matrix; thiol exchange does not contribute meaningfully to SAM loss over the assay window.

**Electrochemical desorption**

Reductive desorption of alkanethiol SAMs on gold occurs via:

$$\text{Au–SR} + e^- \longrightarrow \text{Au}^0 + \text{RS}^- \tag{III.4}$$

and is only possible when the electrode potential reaches or exceeds the desorption potential $E_{des}$. Widrig et al. measured $E_{des}$ for a series of $\omega$-alkanethiol SAMs on gold by cyclic voltammetry in 0.5 M KOH, finding a linear dependence on carbon chain length $n$ [6]:

$$E_{des} = (-0.83 - 0.020\,n)\text{ V vs Ag/AgCl} \quad \text{(0.5 M KOH)} \tag{III.5}$$

For MCH ($n = 6$):

$$E_{des} = -0.83 - 0.020 \times 6 = -0.83 - 0.12 = -0.95\text{ V} \tag{III.6}$$

Love et al. report that desorption of alkanethiol SAMs from gold typically occurs in the range $-1.0 \pm 0.25$ V vs Ag/AgCl [1], giving a conservative lower bound of $-0.75$ V. The SWV scan window runs from $-0.65$ V to $+0.1$ V (Layer 6). The margin between the scan minimum and the conservative lower bound is:

$$\Delta E = E_{des,min} - E_{scan,min} = -0.75\text{ V} - (-0.65\text{ V}) = -0.10\text{ V} \tag{III.7}$$

The scan potential is 100 mV more positive than the most conservative desorption estimate; the Widrig value for MCH is 300 mV more negative than the scan limit. Electrochemical desorption does not occur during the assay.

The following assumptions underlie this section:

1. The oxidative stability argument uses the Schoenfisch $O_3$/air benchmark as an upper bound; no quantitative $H_2O_2$ rate constant is claimed (Limitation 2).
2. Equation (**??**) and the 5.9% bound apply in organic solvent; aqueous exchange is negligible per Schlenoff [7] (Limitation 3).
3. The Love range [1] is used as the primary reference for device-condition $E_{\mathrm{des}}$; the Widrig formula (equation **??**) is supporting data measured specifically in alkaline electrolyte (Limitation 4).

## IV. $Ca^{2+}$ Artefact and Ratiometric Cancellation

**Mechanism**

Wound fluid contains ionised $Ca^{2+}$ at approximately 1.0–1.3 mM (similar to serum). If the probe-thiol architecture includes a PEG spacer between the alkyl chain and the peptide, $Ca^{2+}$ can coordinate the polyether oxygen atoms in a crown-ether-like interaction, contracting the spacer and decreasing the probe-to-gold bridge length by $\Delta d$. Electron transfer (ET) from MB to gold through the bridge follows a tunnelling mechanism, so a decrease in bridge length raises the apparent ET rate constant $k_{\mathrm{ET}}$ and increases the SWV peak current of *uncleaved* probes.

For the peptide cleavage sensor (signal-off: MB released on cleavage), this $Ca^{2+}$-induced increase in $k_{\mathrm{ET}}$ acts on the uncleaved probe population equally before ($I_0$) and after ($I(t)$) the assay. The measured quantity is:

$$\frac{\Delta I}{I_0} = \frac{I_0 - I(t)}{I_0} \tag{IV.1}$$

If $Ca^{2+}$ multiplies $k_{\mathrm{ET}}$ by a factor $R$ for all uncleaved probes identically, then $I_0 \to R\, I_0$ and $I(t) \to R\,(1-f)\, I_0$ where $f$ is the fraction cleaved. Substituting:

$$\frac{\Delta I}{I_0} = \frac{R\, I_0 - R(1-f)\, I_0}{R\, I_0} = f \tag{IV.2}$$

The $Ca^{2+}$ factor $R$ cancels exactly in the signal-off ratio, provided both $I_0$ and $I(t)$ are measured in the same buffer (same $Ca^{2+}$ concentration). The signal is inherently self-normalising against $Ca^{2+}$-induced changes in electron transfer rate, as long as the measurement protocol does not change the buffer between baseline and post-incubation reads.

**Tunnelling decay constant**

The ET rate constant through a molecular bridge of length $d$ is:

$$k_{\mathrm{ET}} = k_{\mathrm{ET}}^{0}\, \exp(-\beta\, d) \tag{IV.3}$$

where $\beta$ is the tunnelling decay constant. Smalley et al. measured $\beta$ for ferrocene-terminated (Fc) alkanethiol chains tethered to gold in 1.0 M $HClO_4$, by fitting the chain-length dependence of the standard rate constant $k^{\circ}$ [8]:

$$\beta = 0.94 \pm 0.06\ \text{Å}^{-1} = (9.4 \pm 0.6) \times 10^{9}\ \text{m}^{-1} \tag{IV.4}$$

This measurement used ferrocene (Fc) as the redox probe, while the present device uses methylene blue (MB). The value of $\beta$ is adopted here because $\beta$ is a property of the alkyl bridge that electrons tunnel through, not of the terminal redox group. Smalley et al. note that $\beta \approx 1.0\ \text{Å}^{-1}$ is consistent across multiple redox probes studied through the same alkyl bridge [8]. The Fc-derived value therefore transfers directly to the MB system.

**Sensitivity of $k_{\mathrm{ET}}$ to bridge contraction**

If $Ca^{2+}$ coordination contracts the PEG spacer by $\Delta d$, the fold increase in $k_{\mathrm{ET}}$ follows from equation (**??**):

$$\frac{k_{\mathrm{ET}}^{\mathrm{Ca}}}{k_{\mathrm{ET}}^{0}} = \exp(\beta\, \Delta d) \tag{IV.5}$$

For a representative contraction of $\Delta d = 0.2$ nm (one carbon–carbon bond displacement, used as a sensitivity bound since $\Delta d$ is not measured for this architecture; see Limitation 5):

$$\frac{k_{\mathrm{ET}}^{\mathrm{Ca}}}{k_{\mathrm{ET}}^{0}} = \exp(9.4 \times 10^{9}\ \mathrm{m}^{-1} \times 0.2 \times 10^{-9}\ \mathrm{m}) \tag{IV.6}$$
$$= \exp(1.88) \approx 6.5$$

Both the active and control electrodes carry the same probe-thiol/MCH SAM and are exposed simultaneously to the same wound fluid, so this fold increase applies identically to both.

**Residual error from probe density change after cleavage**

Equation (**??**) assumes that $Ca^{2+}$ affects uncleaved probes with the same $k_{\mathrm{ET}}$ before and after the assay. This holds exactly if the cleaved probe stub (the fragment remaining on the electrode after MB release) has negligible electron transfer compared with the intact probe. If the stub retains some residual MB (partial cleavage) or if cleavage changes the spacer geometry in a way that modifies the $Ca^{2+}$ response of neighbouring probes, a second-order residual arises. The magnitude of this residual depends on the specific peptide architecture and is not quantifiable without a calibration measurement. It is expected to be small (second-order in $f$ and in $\delta k_{\mathrm{ET,stub}}$), but experimental confirmation that $\Delta I/I_0$ is $Ca^{2+}$-independent in the assay buffer is recommended.

**Limitations of this model**

1. **The deposition protocol and functional coverage limit are now confirmed; the absolute $\Gamma_p$ on the fabricated electrode must still be measured.** Wcisło 2021 [11] provides a confirmed peptide-thiol deposition protocol (5 mM ethanolic solution, 20 h, MCH backfill). Adjémian 2010 [12] confirms the functional coverage regime ($\Gamma_0 \leq 10$ pmol cm$^{-2}$ for >80% cleavage yield with MCH backfill). The device target of 2–5 pmol cm$^{-2}$ is within this regime. What remains unconfirmed is the *actual* $\Gamma_p$ achieved on the fabricated SGKIPRTLTA–MB electrode under the adopted deposition conditions. This must be measured directly by reductive desorption coulometry (method: Widrig [6]) or equivalent before Layer 8 and Layer 9 outputs can be quantified.
2. **No quantitative $k_{\mathrm{ox}}$ for $H_2O_2$ oxidation of thiol SAMs.** Schoenfisch and Pemberton [5] measure SAM oxidation by $O_3$ in air, not by $H_2O_2$ in solution. No paper in `papers/layer_07/` provides $k_{\mathrm{ox}}$ for aqueous $H_2O_2$. The oxidative stability argument is comparative only; experimental confirmation of SAM integrity after 60 min in wound-fluid $H_2O_2$ is required before the assay window can be certified.
3. **Thiol exchange rate confirmed in hexane only.** Schlenoff et al. [7] measure $k_1 = 1.7 \times 10^{-5}$ s$^{-1}$ in hexane and demonstrate negligible aqueous exchange qualitatively; a quantitative rate constant for exchange by Cys or GSH in phosphate-buffered wound fluid at pH 7.4 is not available.
4. **$E_{\mathrm{des}}$ measured in 0.5 M KOH; wound-fluid conditions differ.** The Widrig formula (equation **??**) was determined at pH ≈ 13.7 [6]. At wound-fluid pH 7.4 the Nernst equation predicts that $E_{\mathrm{des}}$ shifts approximately $+(13.7 - 7.4) \times 0.059 \approx +0.37$ V toward positive values (because $RS^-$ is protonated at pH 7.4), placing $E_{\mathrm{des}}$ near $-0.95 + 0.37 = -0.58$ V. This is only 70 mV positive of the scan minimum of −0.65 V, compared with the 300 mV margin in alkaline electrolyte. The Love range [1] ($-1.0 \pm 0.25$ V, conditions unspecified) is used as the primary working reference, but experimental confirmation of $E_{\mathrm{des}}$ in the assay buffer is recommended before the scan window is finalised.
5. **$Ca^{2+}$ self-cancellation in $\Delta I/I_0$ is conditional on the measurement protocol.** Equation (**??**) shows that the $Ca^{2+}$ artefact cancels in $\Delta I/I_0$ only if $I_0$ and $I(t)$ are measured in the same buffer (same $Ca^{2+}$ concentration). If the baseline $I_0$ is measured before adding wound fluid (in calibration buffer without $Ca^{2+}$) and $I(t)$ is measured in wound fluid (with $Ca^{2+}$), the cancellation fails and a spurious signal arises. Protocol design must ensure that both measurements are in the same ionic environment. Whether a PEG spacer is present and its $Ca^{2+}$ affinity are also unconfirmed for this probe architecture.

## References

All papers in `electrochemical_device/papers/layer_07/`.

[1] Love JC, Estroff LA, Kriebel JK, Nuzzo RG, Whitesides GM (2005). Self-assembled monolayers of thiolates on metals as a form of nanotechnology. *Chem Rev* 105(4):1103–1169. **love2005.pdf** – $\Gamma_{\max}$ cross-check (≈ 750 pmol cm$^{-2}$, p. 1108), $E_{\mathrm{des}} = -1.0 \pm 0.25$ V (p. 1122) confirmed from text.

[2] Herne TM, Tarlov MJ (1997). Characterization of DNA probes immobilized on gold surfaces. *J Am Chem Soc* 119(38):8916–8920. **herne1997.pdf** – *DNA-thiol specific. $KH_2PO_4$ buffer requirement, log-time deposition kinetics, and MCH displacement of non-thiolated DNA all pertain to HS-ssDNA with 25 charged phosphate groups per strand. These results do not transfer to the peptide-thiol probe and are not used for deposition protocol in this layer.*

[3] Steel AB, Herne TM, Tarlov MJ (1998). Electrochemical quantitation of DNA immobilized on gold. *Anal Chem* 70(22):4670–4677. **steel1998.pdf** – *DNA-thiol specific. The functional coverage limit ($4 \times 10^{12}$ molecules cm$^{-2}$)*

*was determined by hybridisation steric hindrance. Enzyme-cleavage steric limits for MMP-9 are different and not covered by this paper.*

[4] Petrovykh DY, Kimura-Suda H, Whitman LJ, Tarlov MJ (2003). Quantitative analysis and characterization of DNA immobilized on gold. *J Am Chem Soc* 125(17):5219–5226. **petrovykh2003.pdf** – *DNA-thiol specific. Log-linear adsorption kinetics and the non-Langmuir conclusion apply to HS-ssDNA. Peptide-thiol adsorption kinetics are not covered by this paper.*

[5] Schoenfisch MH, Pemberton JE (1998). Air stability of alkanethiol self-assembled monolayers on silver and gold surfaces. *J Am Chem Soc* 120(18):4502–4513. **schoenfisch1998.pdf** – $O_3$ as primary oxidant confirmed from text; $C_{12}$ SAM 36.9% sulphur loss in 1 h confirmed from Table 5. Paper does not give $k_{ox}$ for aqueous $H_2O_2$; value attributed to this paper in `sam_stability.py` is not confirmed and has been removed.

[6] Widrig CA, Chung C, Porter MD (1991). The electrochemical desorption of *n*-alkanethiol monolayers from polycrystalline Au and Ag electrodes. *J Electroanal Chem* 310(1–2):335–359. **widrig1991.pdf** – $\Gamma_{max} = 9.3 \times 10^{-10}$ mol cm$^{-2}$ (p. 351) confirmed; $E_{des}$ formula (equation **??**) confirmed from text. Measured in 0.5 M KOH; pH correction noted in Limitation 4.

[7] Schlenoff JB, Li M, Ly H (1995). Stability and self-exchange in alkanethiol monolayers. *J Am Chem Soc* 117(50):12528–12536. **schlenoff1995.pdf** – $k_1 = 1.7 \times 10^{-5}$ s$^{-1}$ in hexane confirmed from Table 3; concentration-independence of exchange rate confirmed from text; negligible aqueous exchange confirmed from Figures 3–6. Bimolecular rate model in `sam_stability.py` is not supported by this paper and has been removed.

[8] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu Y-P (2003). The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J Am Chem Soc* 125(7):2004–2013. **smalley2003.pdf** – $\beta = 0.94 \pm 0.06$ Å$^{-1}$ confirmed from text (p. 2009). Measured using ferrocene (Fc) redox probe; value transfers to methylene-blue system because $\beta$ is a property of the alkyl bridge, confirmed from discussion in [8].

[9] Becka AM, Miller CJ (1993). Electrochemistry at $\omega$-hydroxy thiol coated electrodes. 3. Voltage independence of the electron tunneling barrier and measurements of redox kinetics at large overpotentials. *J Phys Chem* 97(23):6233–6239. **becka1993.pdf** – provides context for tunnelling through alkanethiol SAMs; the $\beta = 0.94$ Å$^{-1}$ value used in this layer is from Smalley 2003 [8], not Becka 1993.

[10] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276(23):20572–20578. DOI: 10.1074/jbc.M100900200. **PIIS0021925819405115.pdf** – $k_{cat} = 86.4$ s$^{-1}$, $K_m = 1.3$ mM, $k_{cat}/K_m = 67{,}500$ M$^{-1}$s$^{-1}$ for peptide A11 (SGKIPRTLTA) confirmed from Table II. These solution-phase values are an upper bound for the surface $k_{eff,surf}$.

[11] Wcisło A, Małuch I, Niedziałkowski P, Ossowski T, Prahl A (2021). Label-free electrochemical test of protease interaction with a peptide substrate modified gold electrode. *Chemosensors* 9(8):199. DOI: 10.3390/chemosensors9080199. **10.3390@chemosensors9080199.pdf** – deposition protocol for peptide-thiol on gold (5 mM EtOH, 20 h, backfill with alkanethiol, BSA block) confirmed from Section 2.3.

[12] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing redox peptide monolayers for the rapid measurement of the proteolytic activity of trypsin and $\alpha$-thrombin enzymes. *Langmuir* 26(12):10347–10356. DOI: 10.1021/la100397g. **10.1021@la100397g.pdf** – cleavage yield >80% for $\Gamma_0 \leq 10$ pmol cm$^{-2}$ with MCH backfill confirmed from Fig. 2 and text (p. 10351); working coverage (0.5–1) $\times 10^{-11}$ mol cm$^{-2}$ confirmed. Architecture (Fc-peptide-PEG tethered on Au, MCH backfill) is directly analogous to this device.

[13] Sfragano PS, Moro G, Polo F, Palchetti I (2021). The role of peptides in the design of electrochemical biosensors for clinical diagnostics. *Biosensors* 11(8):246. DOI: 10.3390/bios11080246. **sfragano2021.pdf** – review of peptide-based electrochemical biosensors; confirms PEG linker strategies, cysteine-gold anchoring, and protease cleavage signal-off mechanisms consistent with this device architecture.